



Academic year 2024-2025 (Even Sem)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Date	24 th June 2025	Maximum Marks	(10+50) Marks
Course Code	CD363IA	Duration	(30+90) = 120 Min
Sem	VI Sem	CIE-3	
UG/PG	UG	Faculty: AS	
Data Analytics & Visualization(CD363IA)			

Q. No.	QUIZ	M
1.	Describe the Baseline Models for Value Prediction. In value prediction problems, we are given a collection of feature-value pairs (f_i, v_i) to use to train a function F such that $F(v_i) = v_i$. Baseline models for value prediction problems follow the techniques: Baseline models provide simple yet effective starting points for predicting numeric values. Common approaches include: • Mean or Median Prediction: This model ignores all input features and simply returns the average or median of the target variable. It serves as a benchmark—if a complex model cannot outperform this, the features may be weak or the problem too difficult. • Linear Regression: A straightforward method that fits the best linear relationship between input features and output values. It helps evaluate whether more complex models add meaningful improvement. • Value of the previous point in time: : In time series data, using the most recent past value as the predicted next value is a strong baseline. For instance, predicting tomorrow's temperature based on today's can often be surprisingly effective.	2
2.	Describe Web Analytics with appropriate example. Web Analytics refers to the process of collecting, measuring, analyzing, and reporting web data to understand and optimize website usage and performance. Web Analytics helps organizations track user behavior, page views, session duration, and conversion rates to improve business outcomes. Example: An e-commerce website uses Web Analytics tools like Google Analytics to monitor how many users visit a product page, how long they stay, and what percentage of them make a purchase. This data helps the business improve product placement and user experience.	2
3.	Identify the Data Science Toolkit/packages for the following: a) Data Munging and Wrangling in R b) Data Visualization in R c) Machine Learning Workbench in R d) Google's product for advanced Machine Learning a) Data Munging and Wrangling in R • Packages: dplyr, tidyr, readr, data.table, stringr These packages help in data cleaning, transformation, and manipulation.	2



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	b) Data Visualization in R Packages: ggplot2, lattice, plotly, shiny ggplot2 is the most widely used for static plots, while plotly and shiny support interactive visualization. c) Machine Learning Workbench in R Packages: caret, mlr3, randomForest, xgboost, e1071 caret provides a unified interface for training and tuning models; mlr3 is its modern successor. d) Google's Product for Advanced Machine Learning Tool: TensorFlow TensorFlow is an open-source machine learning framework developed by Google for building and deploying deep learning models.	
4.	Describe the bias-variance trade-off. This conflict between model complexity and predictive performance is captured by the statistical concept known as the bias-variance trade-off. The bias-variance trade-off explains the balance between model simplicity and accuracy. Bias is the error caused by incorrect assumptions in the model (e.g., using a linear model when the true relationship is non-linear), leading to underfitting—poor performance on both training and test data. Variance is the error due to the model being overly sensitive to the training data, especially to noise or fluctuations, leading to overfitting—excellent performance on training data but poor generalization to test data. A good model aims to minimize both bias and variance for better overall accuracy.	2
5.	Write the Block diagram of a basic model evaluation environment. <pre> graph TD Model[Model] <--> Eval[Evaluation Environment] ValData[Validation Data] --> Eval Eval --> Perf[Performance Statistics] Eval --> Err[Error Distributions] Eval --> Conf[Confusion Matrices] </pre>	2



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Q. No.	TEST	M	BT	CO
1.	Write the complete Taxonomy of Models. Illustrate the relevance of these models with suitable real-world examples bringing out the key features and characteristics. Taxonomy of Models- Model types along several different dimensions, reviewing the primary technical issues which arise to distinguish each class. Models have different properties inherent in how they are constructed: ----- 2M • Linear vs. nonlinear • Black box vs. descriptive • First-Principle vs. Data-Driven Models • Stochastic vs. deterministic • Flat vs. hierarchical • Discrete vs. continuous models Illustrate the relevance of these models with suitable real-world examples bringing out the key features and characteristics. ----- 8M Discrete vs. Continuous Models • Discrete models manipulate discrete entities. Representative are discrete-event simulations using randomized (Monte Carlo) methods. • Continuous models forecast numerical quantities over reals. They can employ the full weight of classical mathematics: calculus, algebra, geometry, etc. General vs. Ad Hoc Models • Machine learning models for classification and regression are general, meaning they employ no problem-specific ideas, only specific data. • Ad hoc models are built using domain-specific knowledge to guide their structure and design. • Data science generally seek general models, but ad hoc models can be better.	10	L3	CO5
2.	Apply the complete data science workflow to solve a real-world problem (e.g., such as predicting house prices, student performance, analyzing sales trends). Write the Steps --- 3M Apply the complete data science workflow to solve a real-world problem --- 7M	10	L4	CO1

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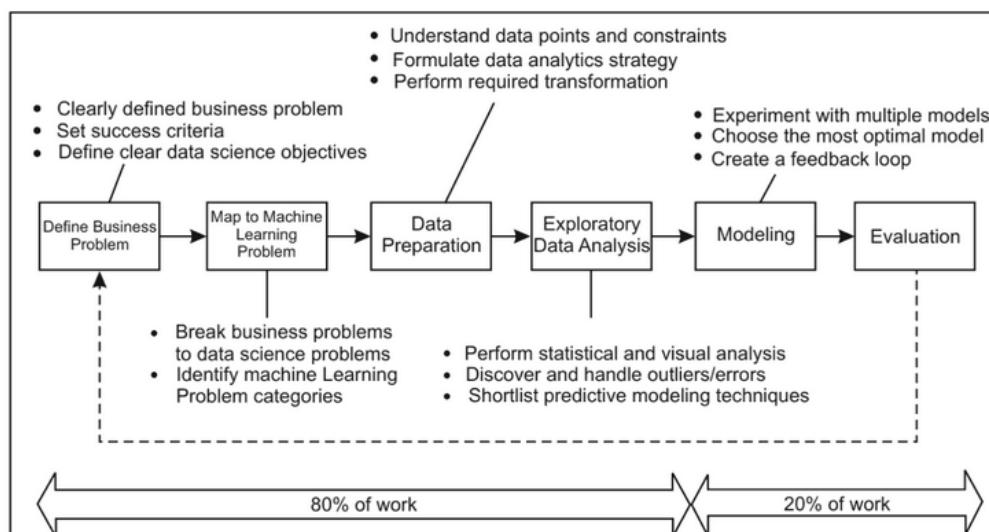


Fig: Seven Steps of Data Science Process

Step 1: Frame or define the (business) problem

Step 2: Collect the raw data needed for your problem (and map it to machine learning in case of Big data)

Step 3: Data Preparation for process the data for analysis

Step 4: Explore the data (Exploratory Data Analysis (EDA))

Step 5: Perform in-depth analysis (Modelling) and producing prescriptive Business Insights

Step 6: Evaluation

Step 7: Visualisation and Communication of Results of the Analyse

Steps in the Data Science Workflow

1. Problem Definition

- Understand the business or research problem clearly.
- Define goals and success metrics.
- Example:* A company wants to reduce customer churn.

2. Data Collection

- Gather data from various sources such as databases, web scraping, IoT sensors, surveys, or logs.
- Example:* Customer demographics, transaction history, support logs.

3. Data Cleaning & Preprocessing

- Handle missing values, remove duplicates, detect and treat outliers.
- Convert data into suitable formats for analysis.
- Example:* Removing incomplete customer records or normalizing income values.

4. Data Exploration (EDA)

- Use descriptive statistics and visualization to understand patterns.
- Identify correlations and trends.
- Example:* Analyzing if churn is related to customer complaints or usage.

5. Feature Engineering

- Create new features or transform existing ones to improve model performance.



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	• <i>Example:</i> Generating features like "average monthly spending" or "complaints per month". 6. Model Building • Apply machine learning or statistical models such as decision trees, regression, clustering, etc. • Select algorithms based on data type and objective. • <i>Example:</i> Using logistic regression to predict if a customer will churn. 7. Model Evaluation • Assess model accuracy using metrics like precision, recall, F1-score, RMSE. • Perform cross-validation. • <i>Example:</i> The churn model gives 85% accuracy and good recall. 8. Deployment • Integrate the model into production (web apps, mobile, dashboards). • Monitor and maintain model performance. • <i>Example:</i> Deploying the churn predictor in the CRM system. 9. Visualization and Communication of Results • Share insights and model outcomes using dashboards, reports, or presentations. • Use visualizations to explain findings to non-technical stakeholders. Real-World Example: Reducing Customer Churn in a Telecom Company 1. Problem: High churn rate affecting revenue. 2. Data Collection: Collect data from CRM and call center logs. 3. Preprocessing: Remove null entries, encode categorical variables. 4. EDA: Discover that churn increases with call complaints. 5. Feature Engineering: Create a variable for "customer complaint frequency." 6. Modeling: Use logistic regression and decision trees. 7. Evaluation: Choose the model with highest AUC score. 8. Deployment: Alert system for high-risk customers is built into CRM. 9. Visualization and Communication: Reports are shared with the customer care team for targeted retention.			
3.	Illustrate how Baseline models for Classification differ from those used in Value Prediction, providing suitable scenarios for each. The first step to assess the complexity of your task involves building baseline models: the simplest reasonable models that produce answers we can compare against. More sophisticated models should do better than baseline models, but verifying that they really do and, if so by how much, puts its performance into the proper context. There are two common tasks for data science models: classification and value prediction. In classification tasks, we are given a small set of possible labels for any given item, like (spam or not spam), (man or woman), or (bicycle, car, or truck). We seek a system that will generate a label accurately describing a particular instance of an email, person, or vehicle. ----- 2M Representative baseline models for classification include: ----- 5M • Uniform or random selection among labels • The most common label appearing in the training data • Somebody else's model --- Same label as the previous point in time. • Clairvoyance --- Rule of thumb heuristics.	10	L3	CO5



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Baseline Models for Value Prediction: In value prediction problems, we are given a collection of feature-value pairs ($f_i; v_i$) to use to train a function F such that $F(v_i) = v_i$. Baseline models for value prediction problems follow from similar techniques to what were proposed for classification, like: ----- **3M**

- Mean or median
- Linear regression:
- Value of the previous point in time

Baseline models must be fair: they should be simple.

Baseline models are deliberately simple “yard-sticks” that let us judge whether more sophisticated algorithms really add value. The two main flavours of supervised learning—**classification** (categorical targets) and **value prediction / regression** (continuous targets). The table and discussion below highlight the key differences and give concrete scenarios you might meet in practice.

1. Purpose of a Baseline

Task type	Why a baseline?
Classification	Show what accuracy you would get by guessing the class in an extremely simple way. If a fancy model cannot beat this, it is probably over-engineered.
Value prediction	Show what error you obtain by returning a trivial numerical estimate. If the RMSE/MAE of an advanced model is not much lower than this, the features or task may be weak.

2. Baseline Models for Classification

Baseline	How it works	Typical scenario
Majority-class (Zero-R)	Always predict the most common class in the training set.	<i>Spam detection:</i> If 80 % of e-mails are “not spam,” a Zero-R classifier attains 80 % accuracy simply by labelling every message “not spam.”
Stratified random / Prior sampling	Predict classes in proportion to their observed frequencies (useful when reporting log-loss).	<i>Credit-default prediction:</i> 5 % of customers default; assign “default” with 5 % probability.
Rule-of-thumb heuristics	Tiny domain-specific rules (e.g., “If age < 25 → likely churn”).	<i>Telecom churn:</i> one-feature splits yield a decision-stump baseline.

Evaluation: Accuracy or macro-F1 is compared against this bar; class-imbalance often makes the majority-class baseline surprisingly tough to beat.



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3. Baseline Models for Value Prediction																	
<table><tr><th>Baseline</th><th>How it works</th><th>Typical scenario</th></tr><tr><td>Mean / Median predictor</td><td>Ignore features and always output the sample mean (or median).</td><td>House-price regression: If the mean sale price is \$300 k, the baseline RMSD is the standard deviation of prices around \$300 k.</td></tr><tr><td>Simple linear regression</td><td>Fit the best straight-line relationship between X and y. Acts as a “first serious model.”</td><td>Predicting fuel efficiency from engine size; a non-linear model should clear this bar.</td></tr><tr><td>Persistence / Last-value (naïve) forecast</td><td>For time series, predict the next value as the last observed one.</td><td>Daily temperature: tomorrow ≈ today. Surprisingly hard to beat for short horizons.</td></tr></table>			Baseline	How it works	Typical scenario	Mean / Median predictor	Ignore features and always output the sample mean (or median).	House-price regression: If the mean sale price is \$300 k, the baseline RMSD is the standard deviation of prices around \$300 k.	Simple linear regression	Fit the best straight-line relationship between X and y. Acts as a “first serious model.”	Predicting fuel efficiency from engine size; a non-linear model should clear this bar.	Persistence / Last-value (naïve) forecast	For time series, predict the next value as the last observed one.	Daily temperature: tomorrow ≈ today. Surprisingly hard to beat for short horizons.			
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Evaluation: Compare Mean Squared Error (MSE), Root Mean Squared Error (RMSE), or Mean Absolute Error (MAE) to these baselines; lower is better.																	
4. Key Differences																	
• Nature of output – Classification deals with discrete labels; baselines revolve around class frequency. Value prediction outputs real numbers; baselines revolve around central tendency or simple functional form. • Metrics – Classification baselines are judged by accuracy/F-score; regression baselines by error magnitudes (MSE, MAE, etc.). • Interpretability – Majority-class and mean predictors are instantly interpretable and data-shape-driven; linear regression gives a first glimpse of feature influence in regression but has no direct analogue in Zero-R classification. • Difficulty to surpass – In highly imbalanced datasets the majority-class baseline can look deceptively “good,” while in noisy numeric data the mean predictor can yield a large error yet still be tough to improve upon.																	
Example Comparison in Practice																	
<table><tr><th>Problem</th><th>Classification baseline</th><th>Regression baseline</th></tr><tr><td>Loan approval (yes/no)</td><td>Always approve (majority); accuracy ≈ 88 %</td><td>Not applicable</td></tr><tr><td>Predict days-on-market for real-estate</td><td>N/A</td><td>Predict average (45 days); RMSE ≈ 12 days</td></tr></table>			Problem	Classification baseline	Regression baseline	Loan approval (yes/no)	Always approve (majority); accuracy ≈ 88 %	Not applicable	Predict days-on-market for real-estate	N/A	Predict average (45 days); RMSE ≈ 12 days						
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4.	A model is built to detect whether a patient has a disease. The model was tested on 300 patients. Out of 120 patients who actually had the disease, 90 were correctly diagnosed. The model also incorrectly predicted 60 healthy individuals as diseased. Using the above information, compute: • Confusion matrix • Accuracy • Precision	10	L4	CO5													



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	- Recall - F1 Score (Assume "Has disease" = Positive class) Soln: - Total patients = 300 - Actual positive (Has disease) = 120 - True Positives (TP) = 90 (correctly diagnosed diseased patients) - False Negatives (FN) = 120 - 90 = 30 (missed diseased patients) - False Positives (FP) = 60 (healthy individuals incorrectly predicted as diseased) - Therefore, True Negatives (TN) = 300 - (TP + FN + FP) = 300 - (90 + 30 + 60) = 120 a) Confusion Matrix -----2M <table border="1"><tr><td></td><td>Actual Diseased (Positive)</td><td>Actual Healthy (Negative)</td></tr><tr><td>Predicted Diseased</td><td>TP = 90</td><td>FP = 60</td></tr><tr><td>Predicted Healthy</td><td>FN = 30</td><td>TN = 120</td></tr></table> b) Accuracy -----2M Accuracy = (TP + TN) / (TP + TN + FP + FN) = (90 + 120) / 300 = 210 / 300 = 0.70 or 70% c) Precision (Positive Predictive Value) -----2M Precision = TP / (TP + FP) = 90 / (90 + 60) = 90 / 150 = 0.60 or 60% d) Recall (Sensitivity or True Positive Rate) -----2M Recall = TP / (TP + FN) = 90 / (90 + 30) = 90 / 120 = 0.75 or 75% e) F1 Score -----2M F1 Score = 2 × (Precision × Recall) / (Precision + Recall) = 2 × (0.60 × 0.75) / (0.60 + 0.75) = 2 × 0.45 / 1.35 = 0.90 / 1.35 ≈ 0.6667 or 66.67%		Actual Diseased (Positive)	Actual Healthy (Negative)	Predicted Diseased	TP = 90	FP = 60	Predicted Healthy	FN = 30	TN = 120			
	Actual Diseased (Positive)	Actual Healthy (Negative)											
Predicted Diseased	TP = 90	FP = 60											
Predicted Healthy	FN = 30	TN = 120											
5a.	What is a Receiver Operating Characteristic (ROC) curve? Explain how it helps in evaluating a classifier's performance. The Receiver Operating Characteristic (ROC) curve provides a visual representation of our complete space of options in putting together a classifier. Each point on this curve represents a particular classifier threshold, defined by its false positive and false negative rates. These rates are in turn defined by the count of errors divided by the total number	04	L3	CO5									

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of positives in the evaluation data, and perhaps multiplied by one hundred to turn into percentages.

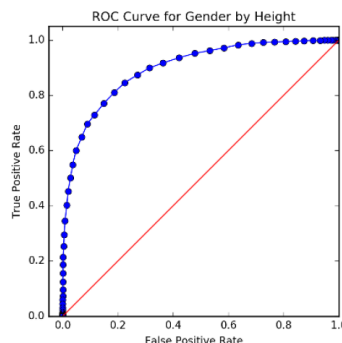
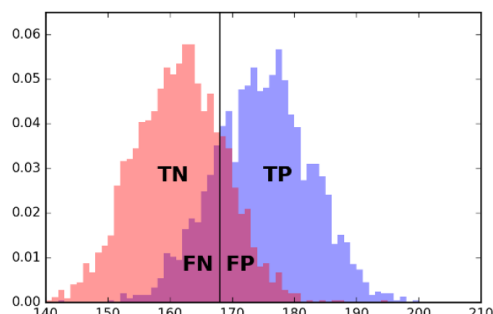


Figure 7.6: The ROC curve helps us select the best threshold to use in a classifier, by displaying the trade-off between true positives and false positive at every possible setting. The monkey ROCs the main diagonal here.

Doing better than the monkey implies an ROC curve that lies above the diagonal. The best possible ROC curve shoots up immediately from (0%,0%) to (0%,100%), meaning it encounters all positive instances before any negative ones. It then steps to the right with each negative example, until it finally reaches the upper right corner.

The area under the ROC curve (AUC) is often used as a statistic measuring the quality of scoring function defining the classifier. The best possible ROC curve has an area of 100% X 100% -> 1, while the monkey's triangle has an area of 1/2. The closer the area is to 1, the better our classification function is.

5b. A regression model is used to predict house prices. For a test set of three houses, the actual prices are [100, 200, 300] (in thousands of dollars) and the predicted prices by the model are [110, 190, 310].

- Calculate the Mean Squared Error (MSE)
- Calculate the Root Mean Squared Error (RMSE)
- Interpret the RMSE in the context of prediction accuracy.

Given:

- Actual values (Y): [100, 200, 300]
- Predicted values ($\hat{Y}$): [110, 190, 310]

Step-by-Step Calculation

Step 1: Find Errors and Squared Errors

-----2M

Observation	Actual (Y_i)	Predicted ($\hat{Y}_i$)	Error ($\hat{Y}_i - Y_i$)	Squared Error ($(\hat{Y}_i - Y_i)^2$)
1	100	110	10	100
2	200	190	-10	100
3	300	310	10	100

a. Mean Squared Error (MSE)

-----1M

$$\begin{aligned} \text{MSE} &= (\sum (\hat{Y}_i - Y_i)^2) / n \\ &= (100 + 100 + 100) / 3 \\ &= 300 / 3 \\ &= 100 \end{aligned}$$

b. Root Mean Squared Error (RMSE)

-----1M



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	$\text{RMSE} = \sqrt{\text{MSE}}$ $= \sqrt{100}$ $= 10$ c. Interpretation of RMSE -----1M The RMSE value of 10 means that, on average, the model's predictions deviate from the actual house prices by \$10,000 (since the units are in thousands of dollars). A lower RMSE would indicate more accurate predictions.			
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